

for output layer having

$$(a_j^L = z_j^L) \quad \left| \quad \begin{aligned} C &= \frac{1}{2} (y - a)^2 \\ \frac{\partial C}{\partial a} &= (y - a)(-1) \end{aligned} \right.$$
$$\frac{\partial a_j^L}{\partial z_j^L} = 1$$

$$\text{output error } \delta_j^L = \frac{\partial C}{\partial a_j^L} \cdot \frac{\partial a_j^L}{\partial z_j^L}$$

$$= -(y - a) \cdot 1$$

$$\delta_j^L = a - y$$

Hence for this type of output layer the quadratic cost fn is directly proportional to the difference

prove:- z_j^L must have form $z_j^L = \ln a_j^L + c$

$$a_j^L = \frac{e^{z_j^L}}{\sum_k e^{z_k^L}}$$

$$\ln(a_j^L) = \ln\left(\frac{e^{z_j^L}}{\sum_k e^{z_k^L}}\right)$$

$$\ln(a_j^L) = \ln(e^{z_j^L}) - \ln\left(\sum_k \cancel{e^{z_k^L}}\right)$$

$$z_j^L = \ln(a_j^L) + c$$

Hence proved.